

Total marks – 120**Attempt Question 1-8****All questions are of equal value**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 1 (15 marks)**Marks**

(a) Find

(i) $\int \frac{\cos \theta}{\sin^5 \theta} d\theta$ **2**

(ii) $\int \frac{dx}{x^2 + 2x + 2}$ **2**

(b) Use the substitution $t = \tan \frac{\theta}{2}$ to find $\int \frac{dx}{5 + 4 \cos x + 3 \sin x}$ **3**

(c) Use the substitution $u = -x$ to evaluate $\int_{-1}^1 \frac{dx}{e^x + 1}$ **3**

(d) Evaluate the following definite integrals:

(i) $\int_0^1 \cos^{-1} x \, dx$ **2**

(ii) $\int_1^2 x(\ln x)^2 \, dx$ **3**

Question 2 (15 marks) Start a new booklet

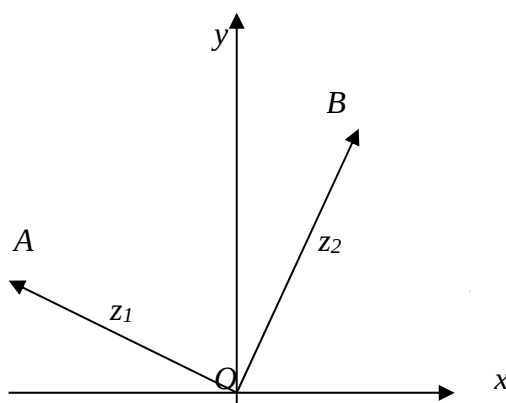
(a) If $z = 3 - 2i$, mark clearly on an Argand diagram the points represented by

(i) $2z$ 1

(ii) $-2iz$ 2

(b) If $|z_1 + z_2| = |z_1 - z_2|$, find the possible values of $\arg\left(\frac{z_1}{z_2}\right)$. 3

(c)



In the Argand diagram, vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ represent the complex numbers

$$z_1 = 2\left(\cos\frac{4\pi}{5} + i\sin\frac{4\pi}{5}\right) \text{ and } z_2 = 2\left(\cos\frac{7\pi}{15} + i\sin\frac{7\pi}{15}\right) \text{ respectively.}$$

(i) Show that $\triangle OAB$ is equilateral 2

(ii) Express $z_2 - z_1$ in modulus-argument form. 3

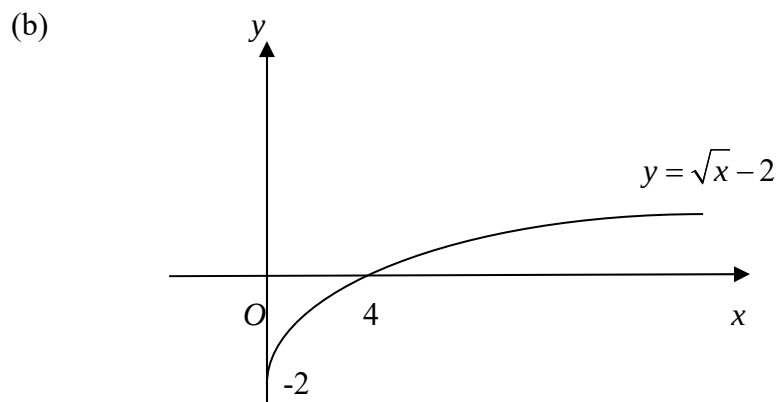
(d) z is a complex number such that $\arg z = \frac{\pi}{3}$ and $|z| \leq 2$.

(i) Show the locus of the point P representing z in the Argand diagram. 2

(ii) Find the possible values of the principal argument of $z - 1$ for z on this locus. 2

Question 3 (15 marks) Start a new booklet

- (a) Twelve different books are made into four parcels of three each. How many different sets of parcels could be made? **3**

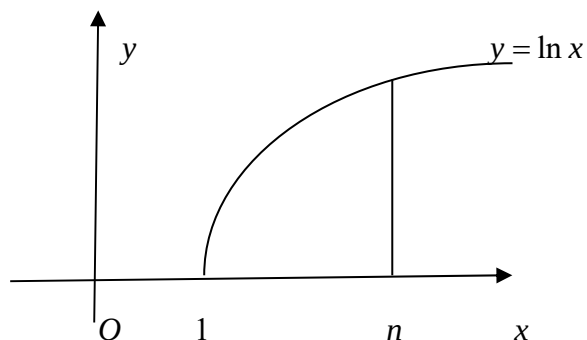


The diagram shows the graph of the function $f(x) = \sqrt{x} - 2$. On separate diagrams sketch the following graphs, showing clearly any intercepts on the coordinate axes and the equations of any asymptotes:

- (i) $y = |f(x)|$ **1**
- (ii) $y = [f(x)]^2$ **1**
- (iii) $y = \frac{1}{f(x)}$ **2**
- (iv) $y = \ln f(x)$ **2**

Question 3 Continued.

(c)



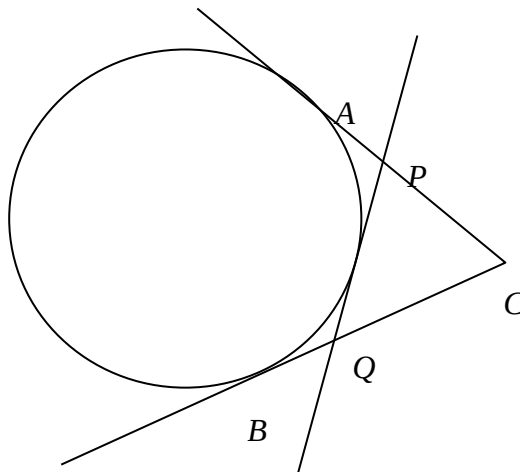
(i) Use the trapezoidal rule with n function values to approximate $\int_1^n \ln x \, dx$ **2**

(ii) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$ and hence find the exact value of $\int_1^n \ln x \, dx$. **2**

(iii) Deduce that $\ln n! < \left(n + \frac{1}{2}\right) \ln n - n + 1$ **2**

Question 4 (15 marks) Start a new booklet

- (a) A and B are two points on a circle. Tangents at A and B meet at C . A third tangent cuts CA and CB in P and Q respectively, as shown in the diagram. **3**
 Show that the perimeter of $\triangle CPQ$ is constant and independent of PQ .



- (b) The polynomial $P(x)$ leaves a remainder of 9 when divided by $(x-2)$ and a remainder of 4 when divided by $(x-3)$. Find the remainder when $P(x)$ is divided by $(x-2)(x-3)$. **4**

(c) $z = \cos \theta + i \sin \theta$

- (i) Show that $z^n + z^{-n} = 2 \cos n\theta$ for $n = 1, 2, 3, \dots$ **2**
 (ii) Hence show that $4 \cos \theta \cos 2\theta \cos 3\theta = 1 + \cos 2\theta + \cos 4\theta + \cos 6\theta$. **3**
 (iii) Hence, solve $\cos^2 \theta + \cos^2 2\theta + \cos^2 3\theta = 1$, giving general solutions. **3**

Question 5 (15 marks) Start a new booklet

(a) $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ are points on different branches of the hyperbola $xy = 9$.

(i) Find the equation of the tangent at P . **2**

(ii) Find the point of intersection, T , of the tangents at P and Q . **2**

(iii) If the chord PQ passes through the point $(0, 2)$, find the locus of T , **3**

(iv) Find the restriction on the locus of T . **1**

(b) The region bounded by the graphs of $y = x^2$ and $y = x + 2$ is revolved around the line $x = 3$. Express the volume of the resulting solid as a definite integral. Do not calculate the value of this integral. **3**

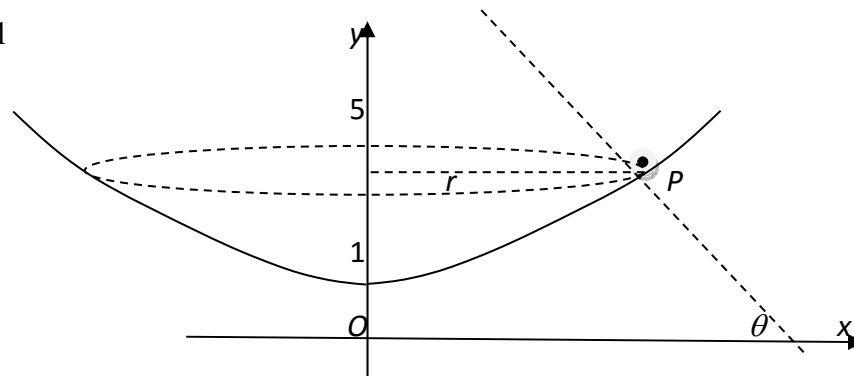
(c) A solid has, as its base, the circular region in the xy -plane bounded by the graph of $x^2 + y^2 = a^2$, where $a > 0$. Find the volume of the solid if every cross-section by a plane perpendicular to the x -axis is an equilateral triangle with one side in the base. **4**

Question 6 (15 marks) [Start a new booklet](#)

- (a) A particle of mass m moves in a straight line away from a fixed point O in the line, such that at time t its displacement from O is x and its velocity is v . At time $t = 0$, $x = 1$ and $v = 0$. Subsequently, the only force acting on the particle is one of magnitude $m \frac{k}{x^2}$, where k is a positive constant in a direction away from O . Show that v cannot exceed $\sqrt{(2k)}$.

(b)

$$y^2 - x^2 = 1$$



A bowl is formed by rotating the hyperbola $y^2 - x^2 = 1$ for $1 \leq y \leq 5$ about the y axis. A particle P of mass m moves around the inner surface of the bowl in a horizontal circle with constant angular velocity ω .

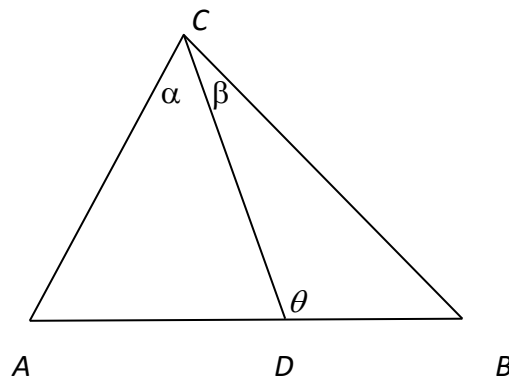
- (i) Show that if the radius of the circle in which P moves is r , then the normal to the surface at P makes an angle θ with the horizontal
where $\tan \theta = \frac{\sqrt{1+r^2}}{r}$. 4
- (ii) Draw a diagram showing the forces acting on P . 1
- (iii) Find expressions for the radius r of the circle of motion and the magnitude of the reaction force between the surface and the particle in terms of m , g and ω . 3
- (iv) Find the values of ω for which the described motion of P is possible. 3

Question 7 (15 marks) Start a new booklet

- (a) The ellipse $\mathcal{E}$: $\left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$ has foci $S(4,0)$ and $S'(-4,0)$. **1**
- (i) Sketch the ellipse $\mathcal{E}$ indicating its foci S, S' and its directrices. **1**
- (ii) Show that the tangent at $P(x_1, y_1)$ on the ellipse $\mathcal{E}$ has equation **1**
 $9xx_1 + 25yy_1 = 225$.
- (iii) The line joining $P(x_1, y_1)$ to $Q(x_2, y_2)$ passes through S . Show that **2**
 $4(y_2 - y_1) = x_1y_2 - x_2y_1$.
- (iv) It is also known that $Q(x_2, y_2)$ lies on $\mathcal{E}$. Show that the tangents at **2**
 P and Q on the ellipse intersect on the directrix corresponding to S .
- (v) Find the equation of the normal to $\mathcal{E}$ at P and decide under what **1**
circumstances, if any, it passes through S or S' .
- (b) $I_n = \int_1^e (1 - \ln x)^n dx, n = 1, 2, 3, \dots$
- (i) Show $I_n = -1 + nI_{n-1}, n = 1, 2, 3, \dots$ **2**
- (ii) Hence evaluate $\int_1^e (1 - \ln x)^3 dx$. **1**
- (iii) Show that $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}, n = 1, 2, 3, \dots$ **2**
- (iv) Show that $0 \leq I_n \leq e - 1$. **1**
- (v) Deduce that $\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{r!} = e$. **2**

Question 8 (15 marks) Start a new booklet

(a)



In $\triangle ABC$, D is the point on AB that divides AB internally in the ratio $m : n$.
 $\angle ACD = \alpha$, $\angle BCD = \beta$ and $\angle CDB = \theta$.

(i) By using the sine rule in each of $\triangle CAD$ and $\triangle CDB$, show that

$$\frac{\sin(\theta + \beta) \sin \alpha}{\sin(\theta - \alpha) \sin \beta} = \frac{m}{n}.$$

(ii) Hence show that $\tan \theta = \frac{(m+n) \tan \alpha \tan \beta}{m \tan \beta - n \tan \alpha}$.

(b) Let $f(x)$ be a function which satisfies the equation:

$$f(xy) = f(x) + f(y) \text{ for all } x, y \neq 0.$$

(i) Show that $f(1) = 0 = f(-1)$ and that $f(x)$ is an even function.

(ii) Prove that $f(x+y) - f(x) = f\left(1 + \frac{y}{x}\right)$ for $x, y, x+y \neq 0$

(iii) Suppose $f(x)$ is differentiable at $x=1$ and $f'(1)=1$. Deduce that

$$f(x) \text{ is differentiable at any } x \neq 0 \text{ and } f'(x) = \frac{1}{x}.$$

End of Examination

Solutions

Question 1

(a) (i) $u = \sin \theta, \quad du = \cos \theta d\theta$

$$\int \frac{du}{u^5} = -\frac{1}{4}u^{-4} + c$$

$$= -\frac{1}{4}\sin^4 \theta + c$$

(ii) $\int \frac{dx}{(x+1)^2 + 1} = \tan^{-1}(x+1) + c$

(b) $t = \tan \frac{x}{2}, \quad \therefore \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} = \frac{1}{2}(1+t^2)$

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\int \frac{\frac{2dt}{1+t^2}}{5 + \frac{4(1-t^2)}{1+t^2} + \frac{6t}{1+t^2}}$$

$$= \int \frac{2dt}{t^2 + 6t + 9} = \int \frac{2dt}{(t+3)^2}$$

$$= 2(t+3)^{-1} + c$$

$$= \frac{-2}{\tan \frac{x}{2} + 3} + c$$

(c) $u = -x, \quad -du = dx$

When $x = -1, u = 1$ and when $x = 1, u = -1$

$$\int_1^{-1} \frac{-du}{e^{-u} + 1} = \int_{-1}^1 \frac{du}{e^{-u} + 1} = \int_{-1}^1 \frac{du}{\frac{1}{e^u} + 1} = \int_{-1}^1 \frac{e^u du}{1 + e^u}$$

$$= \left[\ln(e^u + 1) \right]_{-1}^1 = \ln \frac{e+1}{1/e+1} = \ln e = 1$$

$$(d) \quad (i) \quad \int_0^{\pi/2} \sin y \, dy = \left[-\cos y \right]_0^{\pi/2} = 1$$

$$(ii) \quad \left[\frac{1}{2} x^2 (\ln x)^2 \right]_1^2 - \int_1^2 \frac{1}{2} x^2 \cdot 2 \cdot \frac{1}{x} \ln x \, dx$$

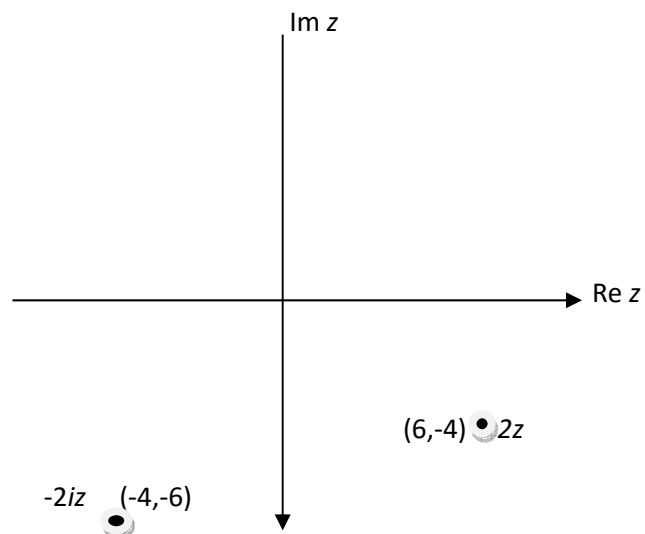
$$= 2(\ln 2)^2 - \int_1^2 x \ln x \, dx$$

$$= 2(\ln 2)^2 - \left[\frac{1}{2} x^2 \cdot \ln x \right]_1^2 + \int_1^2 x \cdot \frac{1}{x} \, dx$$

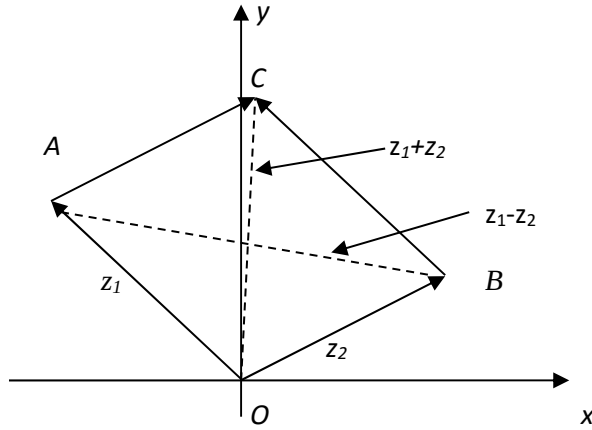
$$= 2(\ln 2)^2 - 2 \ln 2 + 1$$

Question 2

(a)



(b)



Let $\overrightarrow{OA}$, $\overrightarrow{OB}$ represent z_1 , z_2 respectively. Construct the parallelogram $OACB$. Then $\overrightarrow{OC}$, $\overrightarrow{BA}$ represent z_1+z_2 and z_1-z_2 , respectively. Since $|z_1+z_2|=|z_1-z_2|$, $OC=AB$. Hence, $OACB$ is a rectangle.

$$\therefore \angle AOB = \frac{\pi}{2}.$$

But, $\angle AOB = \arg z_1 - \arg z_2$ or $\angle AOB = \arg z_2 - \arg z_1$ (if z_1, z_2 swapped).

$$\therefore \arg \frac{z_1}{z_2} = \pm \frac{\pi}{2}$$

$$(c) \quad (i) \quad \therefore \arg \angle AOB = \arg z_1 - \arg z_2 = \frac{4\pi}{5} - \frac{7\pi}{15} = \frac{\pi}{3}$$

$$OA = OB = 2$$

$$AB = \sqrt{2^2 + 2^2 - 2 \cdot 2 \cdot 2 \cdot \cos \frac{\pi}{3}} = 2$$

$\therefore \triangle OAB$ is equilateral.

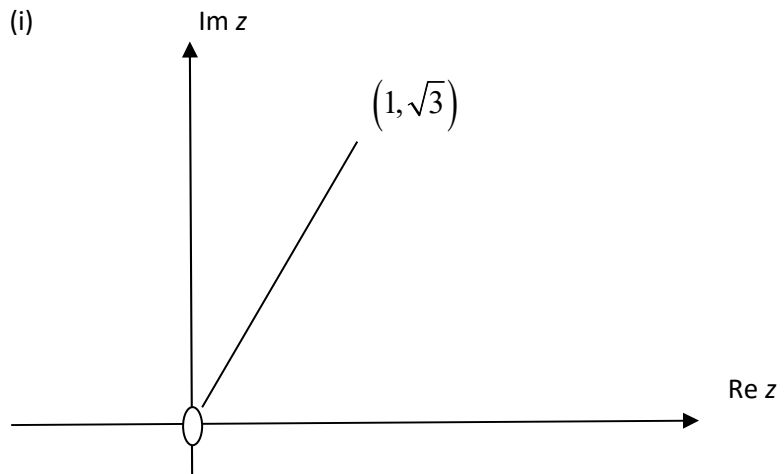
(ii) The vector $\overrightarrow{AB}$ represents $z_2 - z_1$. Now, $\overrightarrow{AB}$ is a clockwise rotation of $\overrightarrow{OB}$ by $\frac{\pi}{3}$.

$$\therefore z_2 - z_1 = z_2 \left(\cos \left(\frac{-\pi}{3} \right) + i \sin \left(\frac{-\pi}{3} \right) \right)$$

$$= 2 \left(\cos \frac{7\pi}{15} + i \sin \frac{7\pi}{15} \right) \left(\cos \left(-\frac{\pi}{3} \right) + i \sin \left(-\frac{\pi}{3} \right) \right)$$

$$= 2 \left(\cos \frac{2\pi}{15} + i \sin \frac{2\pi}{15} \right)$$

(d) (i)

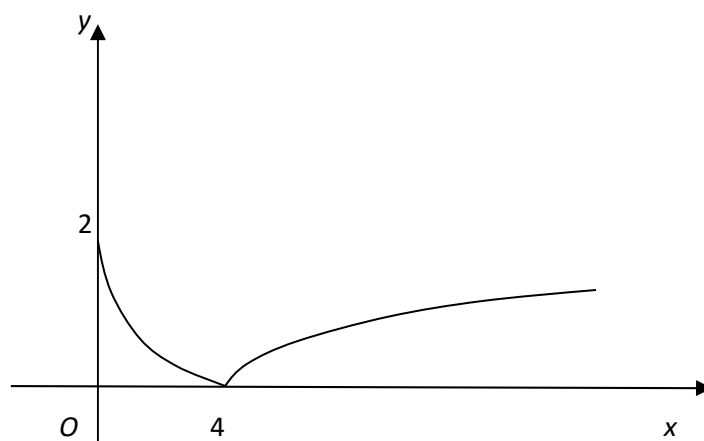


(ii) $\frac{\pi}{2} \leq \arg(z-1) \leq \pi$

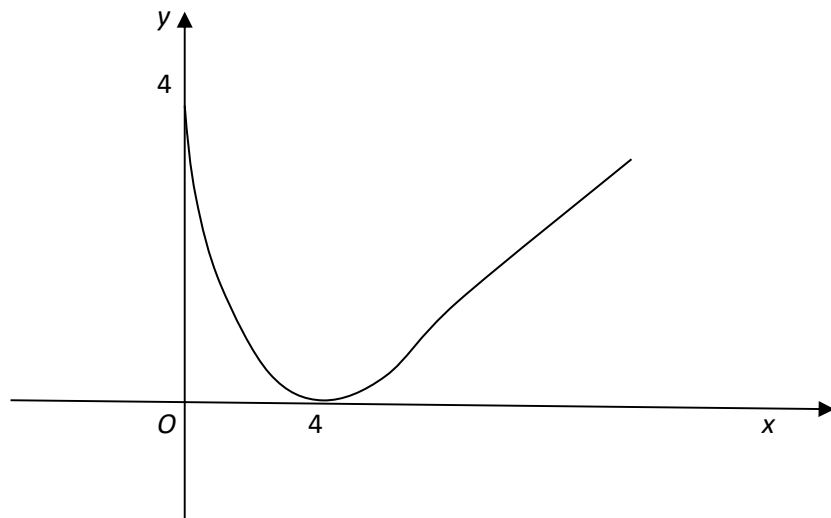
Question 3

(a) $\frac{{}^{12}C_3 \cdot {}^9C_3 \cdot {}^6C_3 \cdot {}^3C_3}{4!} = 15400$

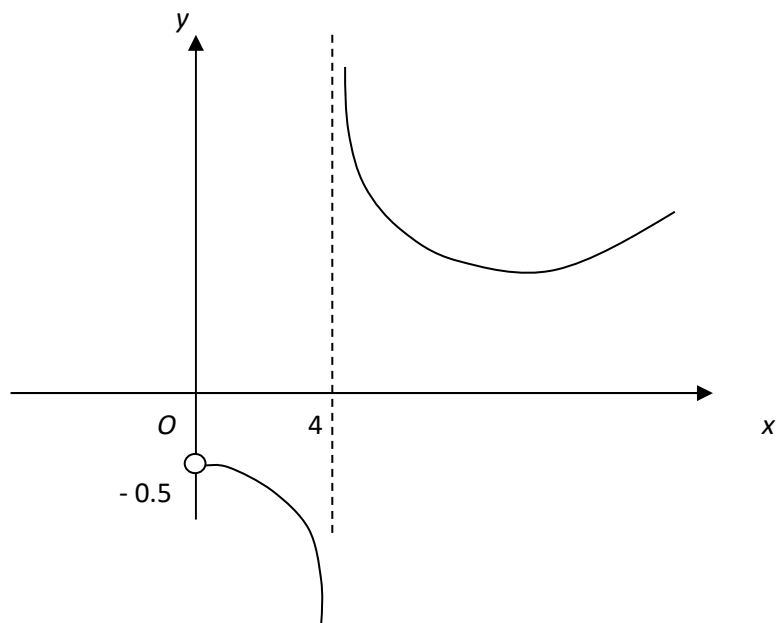
(b) (i) $y = |f(x)|$



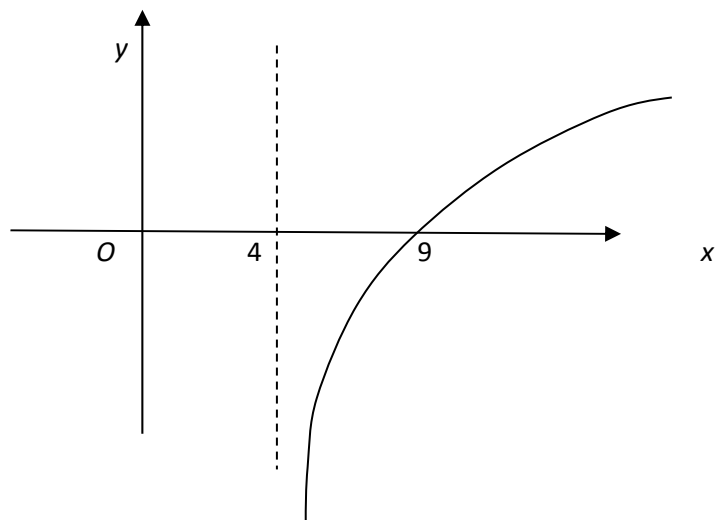
(ii) $y = [f(x)]^2$



(iii) $y = \frac{1}{f(x)}$



(iv) $y = \ln f(x)$



$$(c) \quad (i) \quad \int_1^n \ln x \, dx \approx \frac{1}{2} \left[\ln 1 + \ln n + 2(\ln 2.3.4 \dots (n-1)) \right]$$

$$= \frac{1}{2} \ln n + \ln(1.2.3.4 \dots (n-1))$$

$$= \frac{1}{2} \ln n + \ln(n-1)!$$

$$= \frac{1}{2} \ln n - \ln n + \ln n + \ln(n-1)!$$

$$= \ln n! - \frac{1}{2} \ln n$$

$$(ii) \quad \frac{d}{dx}(x \ln x - x) = x \cdot \frac{1}{x} + \ln x - 1 = \ln x$$

$$\therefore \int_1^n \ln x \, dx = [x \ln x - x]_1^n = n \ln n - n + 1$$

(iii) The trapezia lie below the curve.

$$\therefore n \ln n - n + 1 > \ln n! - \frac{1}{2} \ln n$$

$$\ln n! < n \ln n - n + 1 + \frac{1}{2} \ln n$$

$$\ln n! < \ln n \left(n + \frac{1}{2} \right) - n + 1$$

Question 4

(a) Perimeter of $\triangle CPQ = CP + CQ + PQ$

But $PQ = AP + BQ$ (tangents drawn from P are of equal length and
(tangents drawn from Q are of equal length)

$$\text{Perimeter of } \triangle CPQ = CP + AP + CQ + BQ = CA + CB$$

Which is constant and independent of PQ .

$$(b) \quad P(2) = 9, P(3) = 4$$

$$P(x) = Q(x) \cdot (x-2)(x-3) + R(x), \text{ where } R(x) = ax + b$$

$$P(2) = 0 + 2a + b = 9$$

$$P(3) = 0 + 3a + b = 4$$

$$a = -5, b = 19$$

$$\text{Remainder is } 19 - 5x$$

$$(c) \quad (i) \quad z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \quad (\text{note cosine even and sine odd})$$

$$= 2 \cos n\theta$$

$$(ii) \quad z + z^{-1} = 2 \cos \theta, \quad z^2 + z^{-2} = 2 \cos 2\theta, \quad z^3 + z^{-3} = 2 \cos 3\theta$$

$$8 \cos \theta \cos 2\theta \cos 3\theta = (z + z^{-1})(z^2 + z^{-2})(z^3 + z^{-3})$$

$$= z^6 + 1 + z^2 + z^4 + z^{-4} + z^{-2} + 1 + z^{-6}$$

$$= 2 + z^2 + z^{-2} + z^4 + z^{-4} + z^6 + z^{-6}$$

$$= 2 + 2 \cos 2\theta + 2 \cos 4\theta + 2 \cos 6\theta$$

$$\therefore 4 \cos \theta \cos 2\theta \cos 3\theta = 1 + \cos 2\theta + \cos 4\theta + \cos 6\theta$$

$$(iii) \quad 2 \cos^2 \theta + 2 \cos^2 2\theta + 2 \cos^2 3\theta = 2$$

$$1 + \cos 2\theta + \cos 4\theta + \cos 6\theta = 0$$

$$4 \cos \theta \cos 2\theta \cos 3\theta = 0$$

$$\therefore \theta = 2k\pi \pm \frac{\pi}{2}, \text{ or } \theta = 2k\pi \pm \frac{\pi}{4}, \text{ or } \theta = 2k\pi \pm \frac{\pi}{6}$$

Question 5

(a) (i) $x \frac{dy}{dx} + y = 0, \therefore \frac{dy}{dx} = \frac{-y}{x}$

At P , $\frac{dy}{dx} = \frac{-\cancel{3}/\cancel{p}}{3p} = -\frac{1}{p^2}$

Required equation: $y - \frac{3}{p} = -\frac{1}{p}(x - 3p)$

Which gives $x + p^2 y = 6p$

(ii) tangent at P $x + p^2 y = 6p$

Tangent at Q $x + q^2 y = 6q$

When solved simultaneously, we get the coordinates of T :

$$\left(\frac{6pq}{p+q}, \frac{6}{p+q} \right)$$

(iii) $m_{PQ} = \frac{\cancel{3}/\cancel{p} - \cancel{3}/\cancel{q}}{3p - 3q} = -\frac{1}{pq}$

Equation PQ : $y - \frac{3}{p} = -\frac{1}{pq}(x - 3p)$

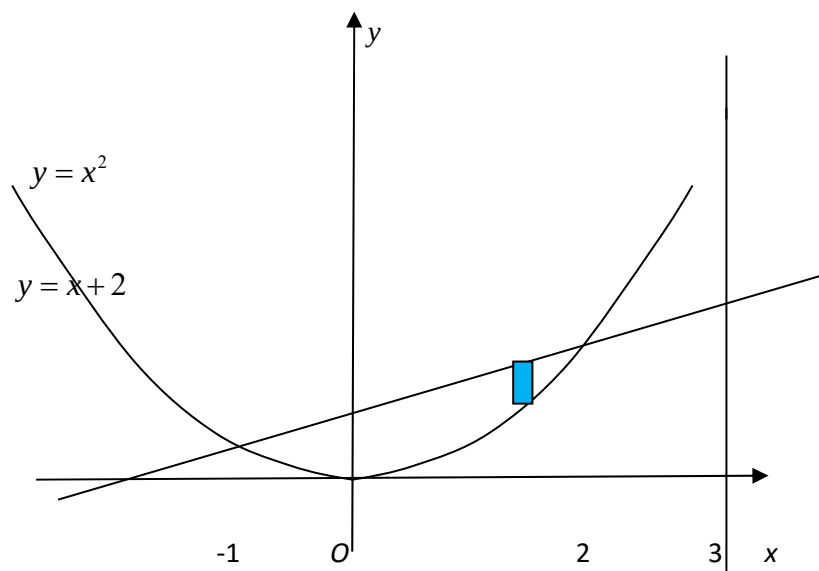
Now, when $x = 0, y = 2$

$$\therefore \frac{p+q}{pq} = \frac{2}{3} \text{ or } p+q = \frac{2pq}{3}$$

At T , $x = \frac{6pq}{p+q} = \frac{6pq}{\cancel{2pq}/\cancel{3}} = 9$

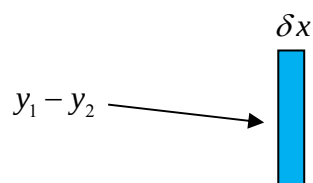
- (iv) Locus of T is $x = 9$, with the restriction that $y < 0$. i.e. T is in the 4th quadrant only.

(b)



To find points of intersection: $x^2 = x + 2, \therefore x = -1, 2$

Consider a typical strip



Rotate the strip to form a shell. The volume of the shell is given by

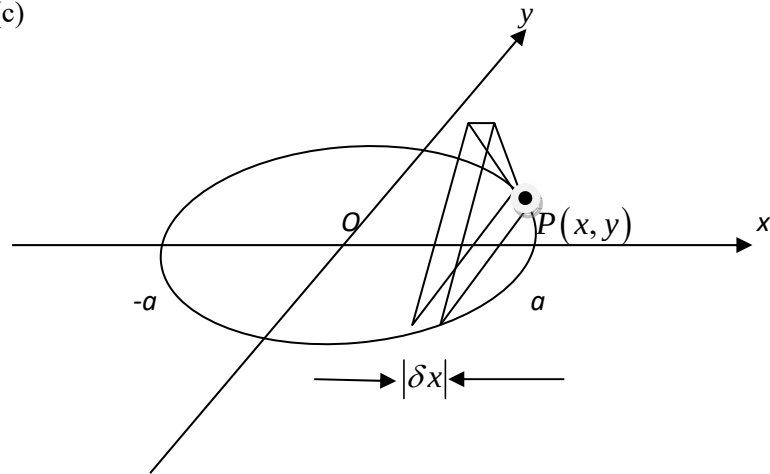
$\delta V = 2\pi r h \cdot \delta x$, where $r = 3 - x$ and $h = y_1 - y_2 = x + 2 - x^2$

$$V \approx \sum_{x=-1}^{x=2} 2\pi(3-x)(x+2-x^2)\delta x$$

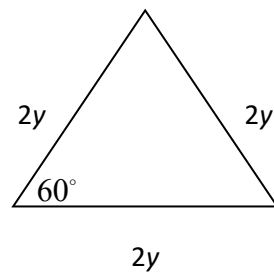
$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-1}^{x=2} 2\pi(3-x)(x+2-x^2)\delta x$$

$$= \int_{-1}^2 2\pi(3-x)(x+2-x^2)dx$$

(c)



Consider a typical slice of width δx .



$$\delta V = \frac{1}{2} 2y \cdot 2y \cdot \sin 60^\circ \cdot \delta x = y^2 \sqrt{3} \cdot \delta x$$

$$V \approx \sum_{x=-a}^{x=a} y^2 \sqrt{3} \cdot \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-a}^{x=a} y^2 \sqrt{3} \cdot \delta x = \lim_{\delta x \rightarrow 0} \sum_{x=-a}^{x=a} (a^2 - x^2) \sqrt{3} \cdot \delta x$$

$$= \sqrt{3} \int_{-a}^a (a^2 - x^2) dx$$

$$= 2\sqrt{3} \int_0^a (a^2 - x^2) dx$$

$$= 2\sqrt{3} \left[a^2 x - \frac{x^3}{3} \right]_0^a$$

$$= \frac{4\sqrt{3}}{3} a^3$$

Question 6

(a) Choose the initial direction as positive

$$\ddot{x} = \frac{k}{x^2}, \quad k > 0$$

$$v \frac{dv}{dx} = \frac{k}{x^2} \Rightarrow v dv = \frac{k}{x^2} dx$$

$$\frac{1}{2} v^2 = -\frac{k}{x} + c, \text{ where } c \text{ is constant}$$

Now, when $x = 1, v = 0 \Rightarrow c = k$

$$\therefore v^2 = 2k \left(1 - \frac{1}{x} \right)$$

$$\text{Now, } x \geq 1 \quad \therefore 0 \leq 1 - \frac{1}{x} < 1$$

$$\therefore 0 \leq v^2 < 2k$$

Hence, v cannot exceed $\sqrt{2k}$

$$(b) (i) \quad y^2 - x^2 = 1 \Rightarrow 2y \frac{dy}{dx} - 2x = 0 \Rightarrow \frac{dy}{dx} = \frac{x}{y}$$

$$\text{At } P, \frac{dy}{dx} = \frac{r}{\sqrt{1+r^2}}$$

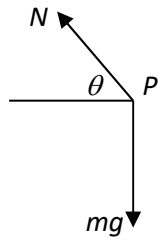
Hence, the gradient of the normal at P is $\frac{-\sqrt{1+r^2}}{r}$

Now, the gradient of the normal is the tangent of the angle made with the 'positive' x axis.

$$\therefore \tan(180^\circ - \theta) = \frac{-\sqrt{1+r^2}}{r}$$

$$\therefore \tan \theta = \frac{\sqrt{1+r^2}}{r}$$

(ii)



(iii) Resolve forces

Horizontally

$$mr\omega^2 = N \cos \theta$$

Vertically

$$mg = N \sin \theta$$

$$\therefore \tan \theta = \frac{g}{r\omega^2} = \frac{\sqrt{1+r^2}}{r} \text{ from part (i)}$$

$$1 + r^2 = \frac{g^2}{\omega^4} \Rightarrow r = \frac{\sqrt{g^2 - \omega^4}}{\omega^2}$$

$$\text{Now, } \cos^2 \theta + \sin^2 \theta = 1$$

$$\therefore \left(\frac{mr\omega^2}{N} \right)^2 + \left(\frac{mg}{N} \right)^2 = 1$$

$$N^2 = m^2 r^2 \omega^4 + m^2 g^2 = m^2 \frac{g^2 - \omega^4}{\omega^4} \omega^4 + m^2 g^2$$

$$= m^2 (2g^2 - \omega^4)$$

$$\therefore N = m\sqrt{(2g^2 - \omega^4)}$$

$$(iv) \quad r = \frac{\sqrt{g^2 - \omega^4}}{\omega^2} \text{ and } r > 0$$

$$\therefore g^2 > \omega^4$$

$$\text{But } N > 0 \quad \therefore 2g^2 > \omega^4$$

Both these conditions exist if $g > \omega^2$

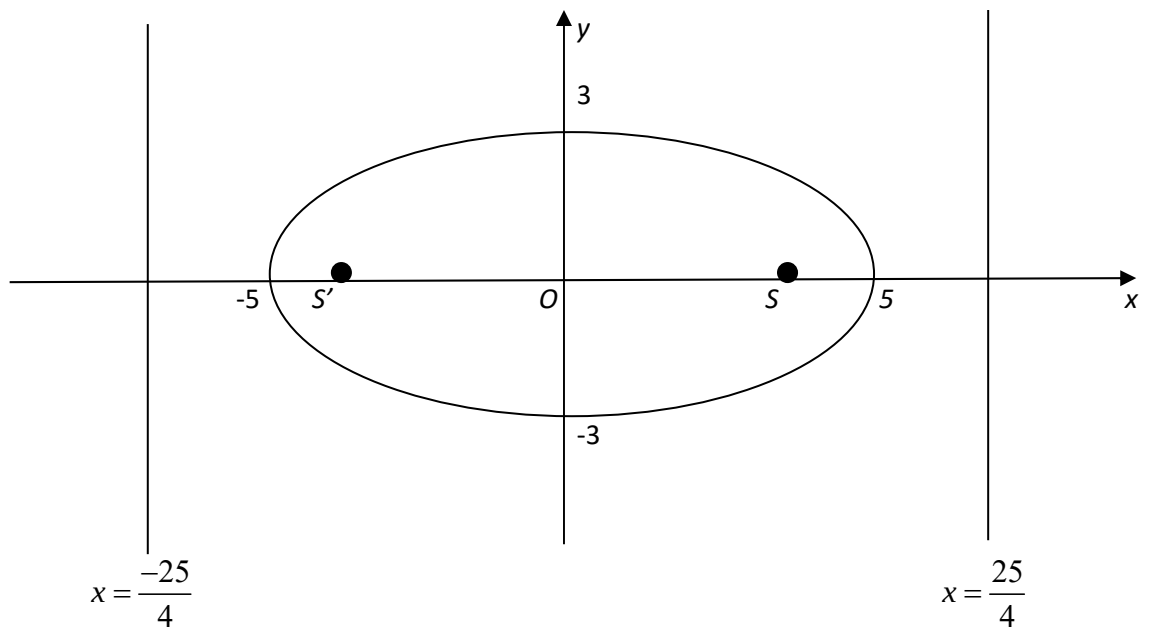
$$\text{Note: } y \leq 5 \Rightarrow y^2 \leq 25 \Rightarrow 1 + r^2 \leq 25$$

$$\therefore \frac{g^2}{\omega^4} > 25 \Rightarrow \omega \geq \sqrt{\frac{g}{5}}$$

$$\therefore \sqrt{\frac{g}{5}} \leq \omega \leq \sqrt{g}$$

Question 7

(a) (i)



$$ae = \pm 4, a = 5, \therefore e = \frac{4}{5}$$

$$\text{Hence, the directrices are } x = \frac{\pm 25}{4}$$

$$(ii) \quad \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\frac{2x}{25} + \frac{2y}{9} \frac{dy}{dx} = 1$$

$$\text{At } P(x_1, y_1), \quad \frac{dy}{dx} = \frac{-9x_1}{25y_1}$$

Required equation:

$$y - y_1 = \frac{-9x_1}{25y_1} (x - x_1)$$

$$9xx_1 + 25yy_1 = 9x_1^2 + 25y_1^2 = 225$$

Note: $P(x_1, y_1)$ lies on the curve $9x_1^2 + 25y_1^2 = 225$

$$(iii) \quad y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$\text{At } x = 4, y = 0$$

$$-y_1(x_2 - x_1) = (y_2 - y_1)(4 - x_1)$$

$$4(y_2 - y_1) = x_1(y_2 - y_1) - y_1(x_2 - x_1) = x_1y_2 - x_2y_1$$

(iv) Two equations are: $9xx_1 + 25yy_1 = 225$ and

$$9xx_2 + 25yy_2 = 225$$

$$25y = \frac{9xx_1 - 225}{y_1} = \frac{9xx_2 - 225}{y_2}$$

$$\therefore 9x(x_1y_2 - x_2y_1) = 225(y_2 - y_1)$$

$$\therefore 9x \cdot 4(y_2 - y_1) = 225(y_2 - y_1)$$

$$x = \frac{225}{36} = \frac{25}{4}$$

$$(v) \quad y - y_1 = \frac{25y_1}{9x_1} (x - x_1)$$

If the normal passes through $S(4, 0)$ then

$$-y_1 = \frac{25y_1}{9x_1}(4 - x_1)$$

$$100y_1 = 16x_1y_1$$

$$\text{Hence, either } x_1 = 6\frac{1}{4} \text{ or } y_1 = 0$$

$$\text{But } -5 \leq x_1 \leq 5, \therefore y_1 = 0$$

$$\text{Similarly, if the normal passes through } S'(-4, 0), y_1 = 0$$

The normals that pass through S and S' are at $(\pm 5, 0)$. The equation of each normal is $y = 0$.

$$\begin{aligned} \text{(b) (i)} \quad I_n &= \int_1^e (1 - \ln x)^n dx \\ &= \left[x(1 - \ln x)^n \right]_1^e - \int_1^e nx(1 - \ln x)^{n-1} \left(-\frac{1}{x} \right) dx \\ &= -1 + nI_{n-1} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad I_3 &= -1 + 3I_2 = -1 + 3(-1 + 2I_1) = -4 - 6(-1 + I_0) \\ &= -10 + 6 \int_1^e dx = -10 + 6(e - 1) = -16 + 6e \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad I_r &= -1 + rI_{r-1} \\ \frac{I_r}{r!} &= \frac{-1}{r!} + \frac{rI_{r-1}}{r!} \\ \frac{I_r}{r!} &= \frac{-1}{r!} + \frac{I_{r-1}}{(r-1)!} \end{aligned}$$

$$\sum_{r=1}^n \frac{I_r}{r!} = \sum_{r=1}^n \frac{-1}{r!} + \sum_{r=1}^n \frac{I_{r-1}}{(r-1)!}$$

$$\frac{I_n}{n!} + \sum_{r=1}^{n-1} \frac{I_r}{r!} = \sum_{r=1}^n \frac{-1}{r!} + \sum_{r=0}^{n-1} \frac{I_r}{r!}$$

$$\frac{I_n}{n!} = \sum_{r=1}^n \frac{-1}{r!} + \frac{I_0}{0!} = \sum_{r=1}^n \frac{-1}{r!} + \frac{1}{0!} = \sum_{r=1}^n \frac{-1}{r!} + \frac{e}{0!} + \frac{-1}{0!} = e - \sum_{r=0}^n \frac{1}{r!}$$

(iv) The domain for the original integrand is $1 \leq x \leq e$

$$\ln 1 \leq \ln x \leq \ln e$$

$$0 \leq \ln x \leq 1$$

$$0 \geq -\ln x \geq -1$$

$$1 \geq 1 - \ln x \geq 1 - 1$$

$$\therefore 0 \leq (1 - \ln x)^n \leq 1$$

$$0 \leq \int_1^e (1 - \ln x)^n dx \leq \int_1^e dx$$

$$0 \leq I_n \leq e - 1$$

(v) $0 \leq \frac{I_n}{n!} \leq \frac{e-1}{n!}$

$$0 \leq \lim_{n \rightarrow \infty} \frac{I_n}{n!} \leq \lim_{n \rightarrow \infty} \frac{e-1}{n!}$$

$$\text{Now, } \lim_{n \rightarrow \infty} \frac{e-1}{n!} = 0, \therefore \lim_{n \rightarrow \infty} \frac{I_n}{n!} = 0$$

$$\therefore \lim_{n \rightarrow \infty} \left(e - \sum_{r=0}^n \frac{1}{r!} \right) = 0, \text{ from part (iii)}$$

$$e - \lim_{n \rightarrow \infty} \left(\sum_{r=0}^n \frac{1}{r!} \right) = 0, \therefore e = \lim_{n \rightarrow \infty} \left(\sum_{r=0}^n \frac{1}{r!} \right)$$

Question 8

(a) (i) $\angle CAD = \theta - \alpha$ (ext $\angle$ of $\triangle CAD$)

$$\therefore \frac{CD}{\sin(\theta - \alpha)} = \frac{AD}{\sin \alpha} \quad (1)$$

$$\angle CBD = 180^\circ - (\theta + \beta) \quad (\angle \text{ sum of } \triangle CBD)$$

$$\therefore \frac{CD}{\sin(180^\circ - (\theta + \beta))} = \frac{DB}{\sin \beta} \quad (2)$$

$$(1) \div (2) \Rightarrow \frac{\sin(\theta + \beta)}{\sin(\theta - \alpha)} = \frac{AD \sin \beta}{DB \sin \alpha}$$

$$\therefore \frac{\sin(\theta + \beta) \sin \alpha}{\sin(\theta - \alpha) \sin \beta} = \frac{AD}{DB} = \frac{m}{n}$$

(ii) $n \sin(\theta + \beta) \sin \alpha = m \sin(\theta - \alpha) \sin \beta$

$$n \sin \alpha (\sin \theta \cos \beta + \cos \theta \sin \beta) = m \sin \beta (\sin \theta \cos \alpha - \cos \theta \sin \alpha)$$

Divide both sides by $\cos \alpha \cos \beta \cos \theta$

$$n \tan \alpha (\tan \theta + \tan \beta) = m \tan \beta (\tan \theta - \tan \alpha)$$

$$(n + m) \tan \alpha \tan \beta = \tan \theta (m \tan \beta - n \tan \alpha)$$

$$\therefore \tan \theta = \frac{(n + m) \tan \alpha \tan \beta}{(m \tan \beta - n \tan \alpha)}$$

(b) (i) $f(x) = f(x \times 1) = f(x) + f(1)$

$$\therefore f(1) = 0$$

$$f(1) = f(-1 \times -1) = f(-1) + f(-1)$$

$$2f(-1) = 0, \therefore f(-1) = 0$$

$$f(-x) = f(-1) + f(x) = f(x)$$

$$\therefore f(x) \text{ is even.}$$

(ii)

$$f(x+y) - f(x) = f\left(x\left(1+\frac{y}{x}\right)\right) - f(x) = f(x) + f\left(1+\frac{y}{x}\right) - f(x) = f\left(1+\frac{y}{x}\right)$$

for $x, y, x+y \neq 0$

$$(iii) \quad f'(1) = 1, \therefore \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = 1$$

$$\therefore \lim_{h \rightarrow 0} \frac{f(1+h)}{h} = 1 \quad (1)$$

Now,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(1+\frac{h}{x}\right)}{h} \quad \text{from (ii)}$$

$$\therefore f'(x) = \frac{1}{x} \lim_{h \rightarrow 0} \frac{f\left(1+\frac{h}{x}\right)}{\frac{h}{x}}$$

Now, as $h \rightarrow 0, \frac{h}{x} \rightarrow 0$ for all $x \neq 0$

Let $u = \frac{h}{x}$.

$$\therefore f'(x) = \frac{1}{x} \lim_{u \rightarrow 0} \frac{f(1+u)}{u} = \frac{1}{x} \quad \text{from (1)}$$

$$2a + 3b + c$$